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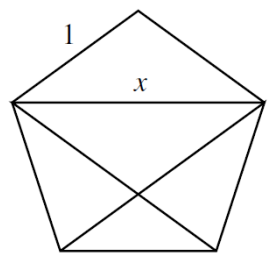
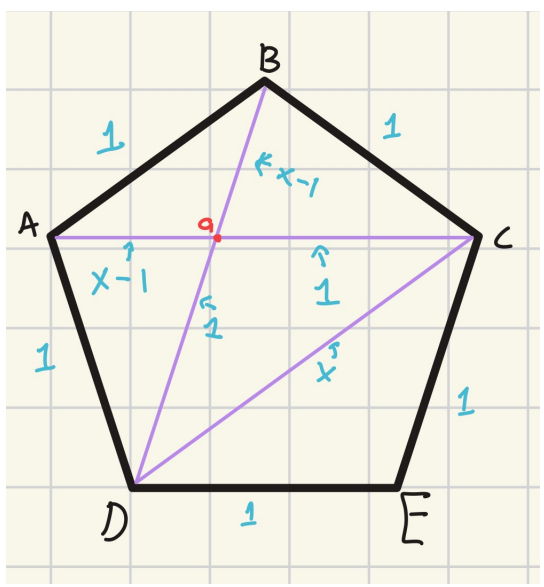
MATH 6400: Quiz #3
Due: January 28, 2026 @ 23:59

1. Assume the figure below is a regular pentagon with each side being length 1. Prove that x is the *Golden Ratio*; that is $x = \frac{1+\sqrt{5}}{2}$. Hint: Find similar triangles and parallel lines to show that x satisfies

$$\frac{x}{1} = \frac{1}{x-1},$$



Solution: We can see since all sides of the pentagon have length 1, this means any line between two non-adjacent corners will have the same length, x. We can then label the points as outlined in the figure below



From here, we can construct diagonals AC and BD, and denote the intersection of these lines a. By transferring the line DE to the line created by aC as described in Euclid I.2, we can see the length of aC is 1. We can then construct the line CD, which as mentioned earlier will have length x.

It becomes apparent aAB forms a triangle similar to aCD, as both form isosceles triangles which are corresponding in their ratios of lengths. We can see $\frac{AB}{aB} = \frac{CD}{aC} \Rightarrow \frac{1}{x-1} = \frac{x}{1}$. Using algebraic manipulation,

we can then solve for x:

$$\begin{aligned}
\frac{x}{1} &= \frac{1}{x-1} \\
(x-1)\frac{x}{1} &= (x-1)\frac{1}{x-1} \\
x(x-1) &= 1 \\
x^2 - x - 1 &= 0 \\
-x^2 + x + 1 &= 0 \\
x &= \frac{-(-1)^2 \pm \sqrt{1-4(-1)(1)}}{2(-1)} && \text{(by \textit{the} quadratic formula)} \\
x &= \frac{1+\sqrt{5}}{2} = \varphi
\end{aligned}$$

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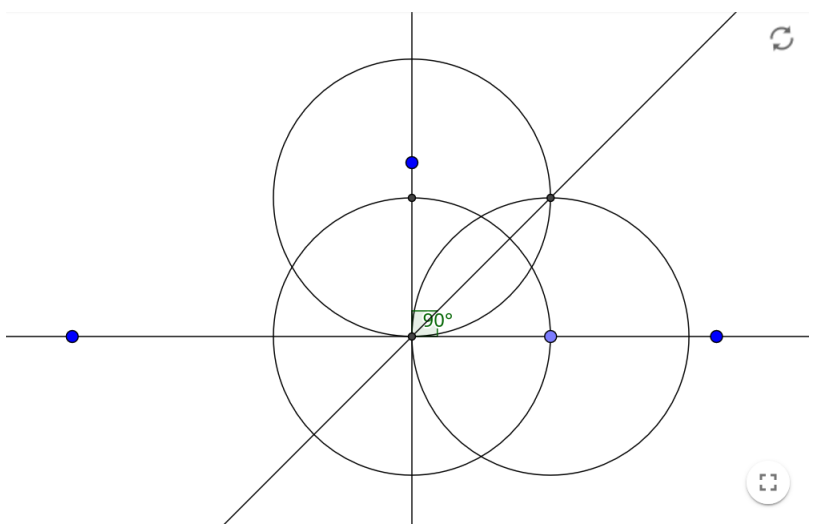
2. Given the result of the first prompt, describe how to construct a regular pentagon using only straightedge and compass. You may assume that φ has been constructed (that is, some length has been constructed that measures φ), and that 'numbers' (distances or lengths of line segments) can be transferred.

   Suggestion: Start by assuming I have the bottom side of the pentagon laid down and that its length is 1. Then use the Golden Ratio (which you may assume has been constructed and is transferable).

3. Trisect a right angle using straightedge and compass. A bullet-proof Euclid-style proof is not necessary, but the construction should be explained enough to be convincing.

   Solution: Below is a construction I performed using Geogebra's *Geometric Constructions* tool, which allows for a digital equivalent of a straight edge and compass.

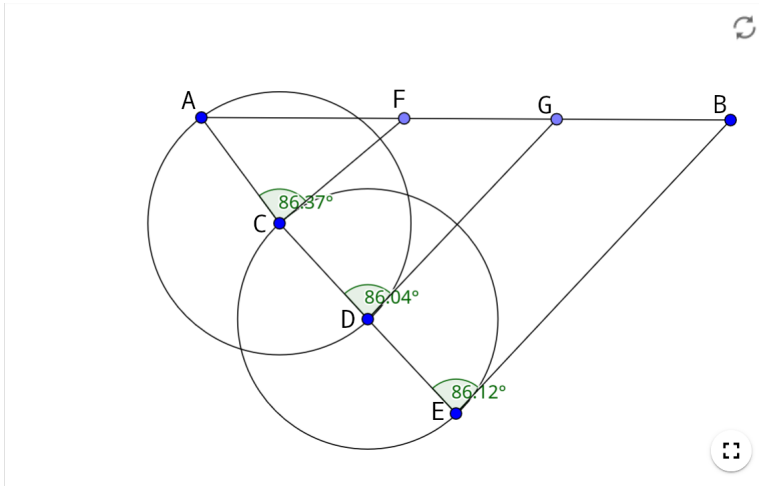


   First I constructed the right angle by creating two tangent lines*. From there, I drew a circle whose center was at the intersection of the lines out some arbitrary distance. From there, I drew two circles whose origin were at the intersection of a line and the circle with a radius with length of the distance between the intersection and the origin. From there, I drew a straight line between the origin and the point of intersection of the circles which did not occur at the origin. This created a perfect bisection of the right angle.

4. Trisect a line segment. For 6400 people: Provide a bullet-proof Euclid-style proof.

   Solution: Once again using Geogebra's *Geometric Constructions* tool, I created this figure using the straightedge and compass

*I am assuming the right angle is given, so I used the angle tool to adjust the lines until the lines were tangent



We label the endpoints of the line segment A and B. From there we draw a line segment out from A and denote the endpoint C. From here, using a compass we draw a circle whose radius is the length AC. We can then construct CD to be the line segment extending the line AC to the edge of the circle. From there we can draw a circle from C towards the point D, and construct the line and point DE and E respectively with the same technique as CD. From here, construct the line BE.

Using the angle of line BE, Construct lines between the points C and D which intersect with the line segment AB; we now have the lines CF and DG. We can then see we have triangles ABE, ACF, ADG which are all similar by construction. Thus, we can see $\frac{AC}{1} = \frac{AE}{3}$ and also $\frac{AD}{1} = \frac{2AE}{3}$. Recall the distance between points A, C, D, E is equal by construction. Since all the triangles ABE, ACF, ADG are similar, then we can see $\frac{AF}{1} = \frac{AB}{3}$ and $\frac{AG}{1} = \frac{2AB}{3}$. Therefore, the line segment is trisected by points F and G.

5. Pretend you are given an arbitrary length (magnitude or *number*) ℓ; using straightedge and compass (anything else would be uncivilized) construct $\sqrt{\ell}$. Provide an argument for your construction; 6400ers: Euclid-style bullet-proof proof.